agree that $U_{\lambda+\rho-\sigma(\rho)} := U_{\langle\lambda+\rho-\sigma(\rho)\rangle}$). Now note that $\mu = \langle\lambda+\rho-\sigma(\rho)\rangle$ is obtained from λ by adding vectors of the form $e_i - e_j$, $i < j$, which implies that $\mu > \lambda$ or $\mu = \lambda$, and the case $\mu = \lambda$ arises only if $\sigma = 1$. [2] Therefore, the above claim follows from Proposition 5.14.1.

Therefore, to show that $\theta_\lambda = \chi_\lambda$, by Lemma 5.7.2, it suffices to show that $(\theta_\lambda, \theta_\lambda) = 1$.

We have

$$(\theta_\lambda, \theta_\lambda) = \frac{1}{n!}\sum_{\mathbf{i}} |C_{\mathbf{i}}|\theta_\lambda(C_{\mathbf{i}})^2.$$

Using that

$$|C_{\mathbf{i}}| = \frac{n!}{\prod_m m^{i_m} i_m!},$$

we conclude that $(\theta_\lambda, \theta_\lambda)$ is the coefficient of $x^{\lambda+\rho}y^{\lambda+\rho}$ in the series $R(x,y) = \Delta(x)\Delta(y)S(x,y)$, where

$$S(x,y) = \sum_{\mathbf{i}}\prod_m \frac{(\sum_j x_j^m)^{i_m}(\sum_k y_k^m)^{i_m}}{m^{i_m} i_m!} = \sum_{\mathbf{i}}\prod_m \frac{(\sum_{j,k} x_j^m y_k^m/m)^{i_m}}{i_m!}.$$

Summing over $\mathbf{i}$ and m, we get

$$\begin{aligned} S(x,y) &= \prod_m \exp(\sum_{j,k} x_j^m y_k^m/m) \\ &= \exp(-\sum_{j,k}\log(1-x_jy_k)) = \prod_{j,k}(1-x_jy_k)^{-1}. \end{aligned}$$

Thus,

$$R(x,y) = \frac{\prod_{i<j}(x_i-x_j)(y_i-y_j)}{\prod_{i,j}(1-x_iy_j)}.$$

Now we need the following lemma.

Lemma 5.15.3.

$$\frac{\prod_{i<j}(z_j-z_i)(y_i-y_j)}{\prod_{i,j}(z_i-y_j)} = \det\left(\frac{1}{z_i-y_j}\right).$$

Proof. Multiply both sides by $\prod_{i,j}(z_i-y_j)$. Then the right-hand side must vanish on the hyperplanes $z_i = z_j$ and $y_i = y_j$ (i.e., must be divisible by $\Delta(z)\Delta(y)$) and is a homogeneous polynomial of degree

[2] Another way to see this is to note that $\sigma(\rho) \le \rho$ lexicographically with equality if and only if $\sigma = 1$, and subtracting both sides from the constant vector $\lambda+\rho$ shows that $\lambda+\rho-\sigma(\rho) \ge \lambda$ with equality if and only if $\sigma = 1$.